

Shared Library — Emerging Technologies

AI / ML • Data Science & Big Data • IoT • Blockchain • other trends | reusable across DSSSB, IBPS & future exams

How to read this shared PDF

This is a **shared library topic**, not tied to one exam. It's referenced by **DSSSB** (Emerging Trends head) and is useful for **IBPS** and essentially any modern CS/IT paper.

Depth is **conceptual/awareness level** — what these technologies are, their key terms, and where they're used. That's what govt/PSU/teaching papers test.

■ **Green box** = how exams test it. ■ **Purple box** = scope note. Security & Cloud have their own full treatment in the IBPS PDF; this one focuses on AI/ML and data-era trends.

0 What this PDF covers

'Emerging technologies' is a recurring head in modern CS/IT syllabi (DSSSB lists it explicitly). Exams test **awareness**: what each technology is, its vocabulary, and typical applications. This PDF gathers them in one reusable place:

- **Part A — Artificial Intelligence & Machine Learning** (the topic you flagged; the heaviest-tested).
- **Part B — Data Science & Big Data** (the 5 Vs, tools).
- **Part C — Internet of Things (IoT)**.
- **Part D — Blockchain & other trends** (AR/VR, robotics, edge, quantum — awareness).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Exams ask **definition + example + vocabulary** MCQs: 'what is supervised learning', 'which is a ML type', 'the 5 Vs of big data', 'what is IoT', 'what is a blockchain'.

For AI/ML the highest-yield split is **supervised vs unsupervised vs reinforcement** learning, and **AI** $\supset$ **ML** $\supset$ **Deep Learning** nesting.

PART A Artificial Intelligence & Machine Learning

A.1 The nesting: AI $\supset$ ML $\supset$ DL

- **Artificial Intelligence (AI)** — the broad goal of making machines perform tasks that need human-like intelligence (reasoning, perception, decision-making).
- **Machine Learning (ML)** — a subset of AI where systems **learn patterns from data** instead of being explicitly programmed.
- **Deep Learning (DL)** — a subset of ML using multi-layer **neural networks**; powers image/speech/language tasks.
- Remember the nesting: **AI $\supset$ ML $\supset$ DL** (each is inside the previous). A very common MCQ.

A.2 Types of machine learning

Type	Learns from	Goal	Examples
Supervised	Labelled data (input→known output)	Predict a label/value	Classification, regression; spam filter
Unsupervised	Unlabelled data	Find hidden structure	Clustering, association; customer segments
Reinforcement	Reward/penalty feedback	Learn best actions by trial	Game-playing, robotics

■ Classification vs regression vs clustering

Classification = predict a *category* (spam/not-spam). **Regression** = predict a *number* (house price). Both are **supervised**. **Clustering** = group similar items with no labels = **unsupervised**. Frequently tested.

A.3 Key concepts & vocabulary

- **Neural network** — layers of connected 'neurons' (input, hidden, output); weights adjusted during training.
- **Training vs testing data** — model learns on training data, is evaluated on unseen test data.
- **Overfitting** — model memorises training data and fails on new data; **underfitting** — too simple to capture the pattern.
- **NLP** (Natural Language Processing) — AI for human language (chatbots, translation). **Computer vision** — AI for images.
- **Generative AI / LLMs** — models that generate text/images (e.g. large language models). Increasingly examined.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Nail: **AI $\supset$ ML $\supset$ DL**; the **three learning types** + one example each; **classification vs regression vs clustering**; **overfitting** definition; **NLP/CV** meaning.

PART B Data Science & Big Data

B.1 Data Science

Data science extracts insight from data by combining statistics, programming, and domain knowledge. Typical pipeline: **collect** → **clean** → **explore** → **model** → **visualise** → **decide**. Tools: Python (Pandas/NumPy/scikit-learn), R, SQL, Tableau.

B.2 Big Data — the 5 Vs

V	Meaning
Volume	Huge amounts of data (TB/PB and beyond)
Velocity	Data generated & processed at high speed (real-time streams)
Variety	Many formats — structured, semi-structured, unstructured
Veracity	Uncertainty / trustworthiness of the data
Value	Usefulness of insight extracted

- **Structured** (tables/DB) vs **semi-structured** (JSON/XML) vs **unstructured** (text, images, video) data.
- **Hadoop** — framework for distributed storage (**HDFS**) & processing (**MapReduce**) of big data. **Spark** — faster in-memory alternative.
- Big data links to cloud (storage/compute) and ML (analytics).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

High-yield: **the 5 Vs** (Volume/Velocity/Variety/Veracity/Value), **structured vs unstructured**, and **Hadoop/MapReduce/HDFS** as the big-data stack.

PART C Internet of Things (IoT)

- **IoT** — a network of physical objects (**things**) embedded with **sensors**, software, and connectivity that collect and exchange data over the internet.
- **Architecture layers**: perception (sensors/actuators) → network (connectivity) → processing (cloud/edge) → application (user services).
- **Sensors** (measure — temperature, motion) vs **actuators** (act — motors, switches).
- **Edge computing** — processing data near the source (the device) instead of sending everything to the cloud — reduces latency.
- Applications: smart homes, smart cities, wearables, industrial IoT, agriculture.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

MCQs: **what IoT is**, **sensor vs actuator**, **edge computing** purpose (low latency, near-source), and IoT **application examples** (smart city/home).

PART D Blockchain & Other Trends

D.1 Blockchain

- **Blockchain** — a **distributed, immutable ledger**: data is stored in **blocks** chained by cryptographic **hashes**, replicated across many nodes (decentralised).
- Key properties: **decentralisation** (no single authority), **immutability** (past blocks can't be altered), **transparency**, and consensus (e.g. Proof of Work).
- Uses: cryptocurrencies (Bitcoin), smart contracts (Ethereum), supply-chain tracking. Links to cryptography & hashing (see IBPS security PDF).

D.2 Other awareness-level trends

Technology	One-liner
AR / VR	Augmented Reality (overlay on real world) / Virtual Reality (fully simulated environment)
Robotics	Machines that sense, plan, and act physically; often AI-driven
Quantum computing	Uses qubits (superposition/entanglement) for certain problems far faster
Edge / Fog computing	Processing near the data source rather than central cloud
5G	Ultra-fast, low-latency mobile networks enabling IoT/AR at scale
Digital twin	A live virtual replica of a physical system for simulation/monitoring

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Blockchain MCQs: **distributed immutable ledger**, **decentralisation**, blocks linked by **hashes**. Others are pure one-line recall: **AR vs VR**, **qubit**, **edge computing**.

■ Coverage note / warning

This whole PDF is **awareness level** by design — govt/PSU/teaching exams test recognition and vocabulary, not implementation. Don't go deeper unless a specific future exam (e.g. a data-science-focused one) demands it; then it becomes its own library module.

Rapid recap — emerging tech (reusable)

Area	Nail these
AI/ML nesting	AI $\supset$ ML $\supset$ Deep Learning; ML learns patterns from data; DL uses neural networks
ML types	Supervised(labelled→predict), Unsupervised(unlabelled→cluster), Reinforcement(reward/trial)
ML tasks	Classification(category) & Regression(number) = supervised; Clustering = unsupervised; overfitting; NLP/CV
Big Data	5 Vs: Volume/Velocity/Variety/Veracity/Value; structured vs unstructured; Hadoop/HDFS/MapReduce, Spark
IoT	Things+sensors+connectivity; sensor vs actuator; edge computing (near-source, low latency); smart city/home
Blockchain	Distributed immutable ledger; blocks linked by hashes; decentralised; Bitcoin/smart contracts
Other trends	AR(overlay) vs VR(simulated); robotics; quantum(qubits); 5G; digital twin

✓ Where this sits

This is a **shared library topic** covering the emerging-tech head for **DSSSB** and useful for **IBPS** + future exams. Combined with the umbrella, the three exam-wise PDFs, and the IBPS Cloud/Security PDF, your **CS/IT coverage for all three target exams is complete**.

Remaining plan items are pure future-proofing: optional **library-filler** modules (Distributed Systems, deeper Web, standalone Cryptography, deeper AI/ML) for exams you haven't committed to yet, and the **Master Topic Index** switchboard. Q&A sets follow your walkthroughs.